

Plan — renewal invoicing for service agreements

29 September 2026. Status: plan, waiting for go-ahead. Builds on migration 110 (agreements per bike) and SERVICE-AGREEMENTS-HANDLING-2026-09 §6. Nothing here is built.

What it must do

Each covered bike renews on the anniversary of its start date (its line's `start_date`), **invoiced 12 months ahead, a month before the anniversary**, and renews unless the line has ended. Today Dennis does this by hand from the register's twelve month sheets. The real thing looks like e-conomic invoice 7114: - product JP-SERVFL "Forlængelse af serviceaftale(r)"; - the frame numbers and the 12-month period on the invoice; - 1 704 kr per bike; - the **department's EAN**, and 30 days to pay.

The rule: - **When:** on the 1st of each month, the system drafts the renewals for every anniversary in the **next** month. That is Dennis's month-sheet rhythm, so December's anniversaries are drafted on 1 November. - **How it is grouped:** one draft invoice per customer × department EAN, one line per bike, each line carrying its own period (anniversary → day before the next). - **Price:** the line's frozen yearly price. - **Payment:** 30 days. - **Who acts:** Dennis reviews, issues and sends. Nothing is issued by itself.

Decisions needed before building

1. **How does a renewal reach a municipality? (owner, planning chat)** This is the one real conflict.
 - Public customers must receive e-invoices by EAN (NemHandel). Today e-conomic sends them, because Dennis makes them there as invoices.
 - The FMS pushes issued invoices to e-conomic as **journal vouchers** (DECISIONS 2026-07-09: the FMS owns the INV number series). A voucher is a booking, not a document, so e-conomic cannot send it by EAN.
 - The same gap exists for every FMS invoice to a public customer. Renewals only make it urgent, because they are most of them.
 - Options:
 - **a) Renewals go to e-conomic as draft INVOICES** (product JP-SERVFL, the department's EAN), and e-conomic books and sends them. E-conomic's number is then the customer-facing one for this kind, and the FMS keeps its draft as the record. Cheapest route to "sent by EAN", but it forks the number series for one invoice kind.
 - **b) The FMS sends OIOUBL itself** through a NemHandel access point (a provider, a registration, a new outbound adapter). Keeps one number series; clearly the biggest job.
 - **c) The FMS drafts, and Dennis re-keys in e-conomic**, as today but from a list. No integration; the manual step stays.
 - **Recommendation:** start with (c), because the list alone removes the spreadsheet work. Decide (a) or (b) before 1 January. Phase A below is the same whichever is chosen.
2. **The switch-over month (Dennis).** From which month the system drafts instead of the spreadsheet. Until it is set, the job does nothing, so the same month is never invoiced twice.
3. **Retire the monthly-fee engine now (owner).** Recommended. It bills `service_agreements.monthly_fee` monthly in arrears, which DECISIONS 2026-09-29 already says

renewals replace, and production has no fee invoices. Keeping both means two engines that can each bill the same bike.

4. **Bikes with no EAN (Dennis).** 141 register rows have none. Recommended: invoice them to the customer's own EAN and list them for him.
5. **The invoice text (Dennis).** Recommended: copy invoice 7114 word for word, with the frame number and the period on each line.

Phase A — buildable now

Data (migration 111) - `invoices.kind` gains `renewal`. - `invoices.organization_unit_id`: the department the invoice is for. At issue the EAN becomes `COALESCE(unit.ean, org.ean)`, for **all** invoices; today only the customer's EAN is ever used, and nothing reads the department's. - `invoices.payment_terms_days`: frozen on the draft (30 for renewals). The due date at issue = issue date + this, else the customer's terms. - `invoice_lines.service_agreement_bike_id`, next to the existing `service_agreement_id` + `billing_period_start/end`. - `app_settings.renewals_from` (the switch-over date; NULL means the job is off) and `renewal_payment_terms_days` (30). Operational config goes on `/admin/settings`, config doctrine tier 2. - An RPC `insert_renewal_invoice(payload)` writes one draft and its lines in one transaction. Under a lock it re-checks that no line is already billed for that period, so the job and *Run now* running at the same moment cannot double-bill.

The engine - A pure planner, `src/lib/service-agreements/renewals.ts`. For each active line it works out the next anniversary after what is already billed, and keeps the ones that: - fall in the target month; - fall on or after `renewals_from`; - are not ended before the anniversary.

It then groups them into customer × EAN drafts. - **"Already billed" = the latest `billing_period_end` for that bike line on a live invoice.** Cancelled and credited invoices and credit-note mirror lines are ignored, the same guard the fee engine uses. Cancelling a draft therefore frees its bikes to be drafted again. - **Never drafted, but listed for Dennis:** - a line with no yearly price (the backfilled ones, and the register's blank-price rows); - a bike with no EAN anywhere; - a price of 0 kr. - **The drafting code,** `src/lib/invoicing/renewal-drafts.ts`, takes a Supabase client and is shared by the button and the job, so it can run from cron. The current fee action cannot: it uses the session client.

Running it - A daily job `draft-renewals` (JOBS entry, a thin route, a `vercel.json` line). It acts only on the 1st, and it is safe to run twice. - *Run now* on `/admin/jobs`, and *Draft renewals now* on `/invoices`. - A notice to Dennis, `renewals.drafted`: "N renewal drafts for December, M bikes, X kr; K bikes need a price or an EAN".

Screens - `/invoices` → a **Renewals** section, replacing the fee section: - next month's due renewals per customer × EAN (bikes, total); - the drafts made; - the problem list. - **The invoice page and print** show the department, the EAN used and each line's period, with the heading "Forlængelse af serviceaftale". - **The agreement page:** each bike line shows *renewed until* and *next renewal*. - **The customer page:** a department's EAN becomes editable in the departments section. Today nothing edits `organization_units.ean_number`. - **Dashboard:** "uninvoiced" counts renewals due next month instead of unbilled monthly fees.

Changing a bike after a draft exists: end the line, cancel the draft, then run again. Draft invoice lines stay non-editable, as they are today.

Retired (if decision 3 is yes): - `agreement-fees.ts`, `create-agreement-fee-invoices.ts` and the fee button; - `findAgreementMonthlyFees`; - the *Monthly fee* field and column on agreements.

The columns stay, so the change can be reversed.

Checks - A new invariant in `scripts/audit-invariants.sql` : no bike line on two live invoices with overlapping periods. - `npm run smoke` . - Local run-through: - draft a month; - run it twice, which must add nothing; - cancel a draft and re-run; - end a line and re-run; - issue a draft and check its due date and EAN.

Phase B — delivery (after decision 1)

- (a) or (b) from decision 1.
- Whichever it is: the e-conomic push must use the customer numbers that **already exist** in e-conomic (`external_customer_no` on the customer and on the department). Today it ignores them and creates a new customer on first push, which would duplicate every existing customer, and there is no customer per department. This has to be fixed before any production push, renewals or not.
- The production e-conomic grant (still trial).

Phase C — the register's agreements (separate)

The import in the handling document §4. Two points matter here: - every imported line needs a **renewed-until** date, taken from the register (the last year invoiced); - `renewals_from` must be at least the switch-over month.

Without both, the first run would draft years that Dennis already invoiced by hand.

Files (phase A)

AREA	FILES
Migration	migrations/111_renewal_invoicing.sql
Engine	src/lib/service-agreements/renewals.ts (pure), src/lib/invoicing/renewal-drafts.ts , src/lib/invoicing/settings.ts
Job + notice	src/lib/cron/jobs.ts , src/app/api/cron/draft-renewals/route.ts , vercel.json , src/lib/people/notifications.ts , src/lib/people/email-content.ts
Invoices	src/app/invoices/page.tsx , _components/renewals-section.tsx , _actions/draft-renewals.ts , _actions/transition-invoice.ts , [id]/page.tsx , [id]/print/page.tsx
Agreements	src/app/service-agreements/[id]/_components/agreement-bikes-panel.tsx + its page
Customers	src/app/organizations/[id]/_components/units-section.tsx (+ its dialog and action)
Settings	the /admin/settings invoicing card
Dashboard	src/lib/dashboard/queries.ts , src/app/page.tsx
Retire	src/lib/invoicing/agreement-fees.ts , src/app/invoices/_actions/create-agreement-fee-invoices.ts , _components/draft-fee-invoices-button.tsx , src/lib/invoicing/uninvoiced.ts , the agreement form + list
Other	scripts/audit-invariants.sql , messages/{en,da}.json , src/lib/types/database.ts , DECISIONS + CLAUDE.md

Estimate, phase A: ~600 human-dev-min (~60 min wait).